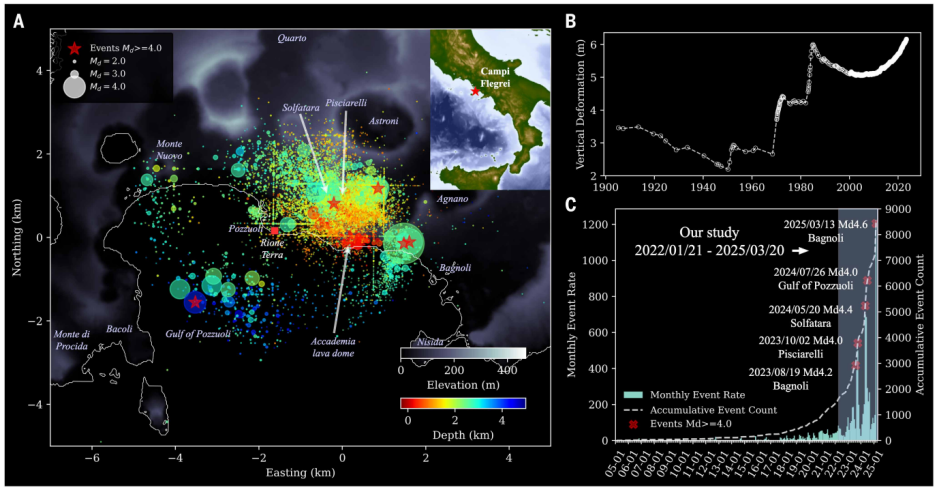
Figure 1: Seismicity and ground displacement

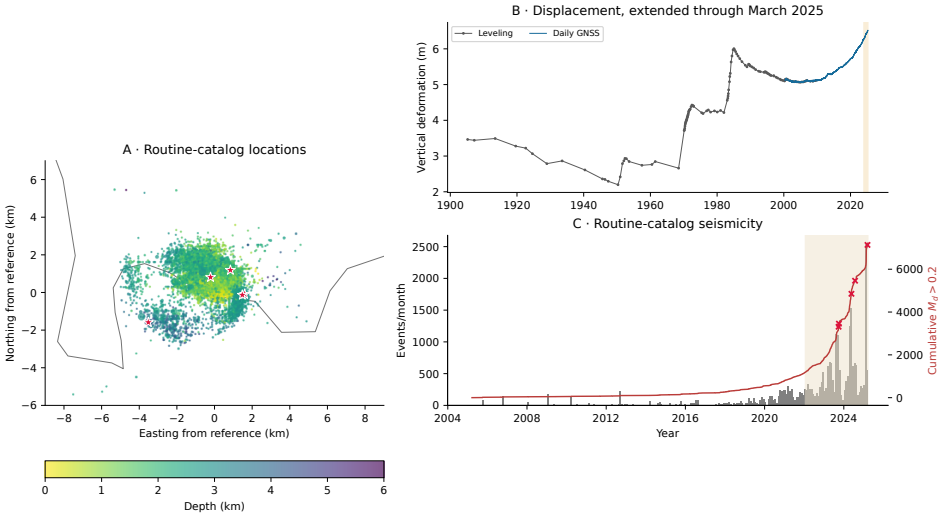
Published reference



Tan et al. (2025), tanmodel.pdf, p. 2.
DOI: 10.1126/science.adw9038.

Independent result

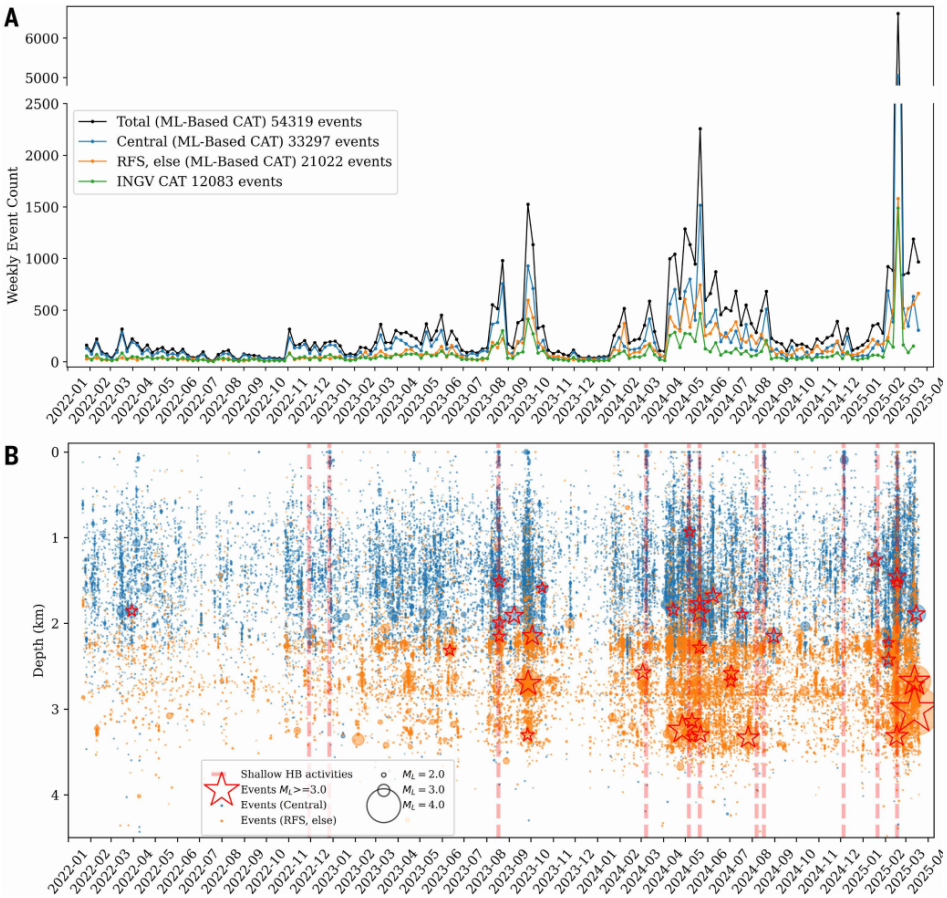
Panel 1A–C — partially reproduced



Partial reconstruction from permitted original displacement observations and external routine-catalog records. A: 16,046 located routine events and five Md>=4 stars. B: historical leveling datum; daily GNSS through 20 March 2025, with the extension beyond 2023 shaded. C: all monthly counts and exact-time cumulative Md>0.2; March includes days 1–20 only. This is not the independently detected catalog. The basemap and live catalog snapshot differ from the publication.

Figure 2: Temporal evolution of seismicity

Published reference



Tan et al. (2025), tanmodel.pdf, p. 3.

DOI: 10.1126/science.adw9038.

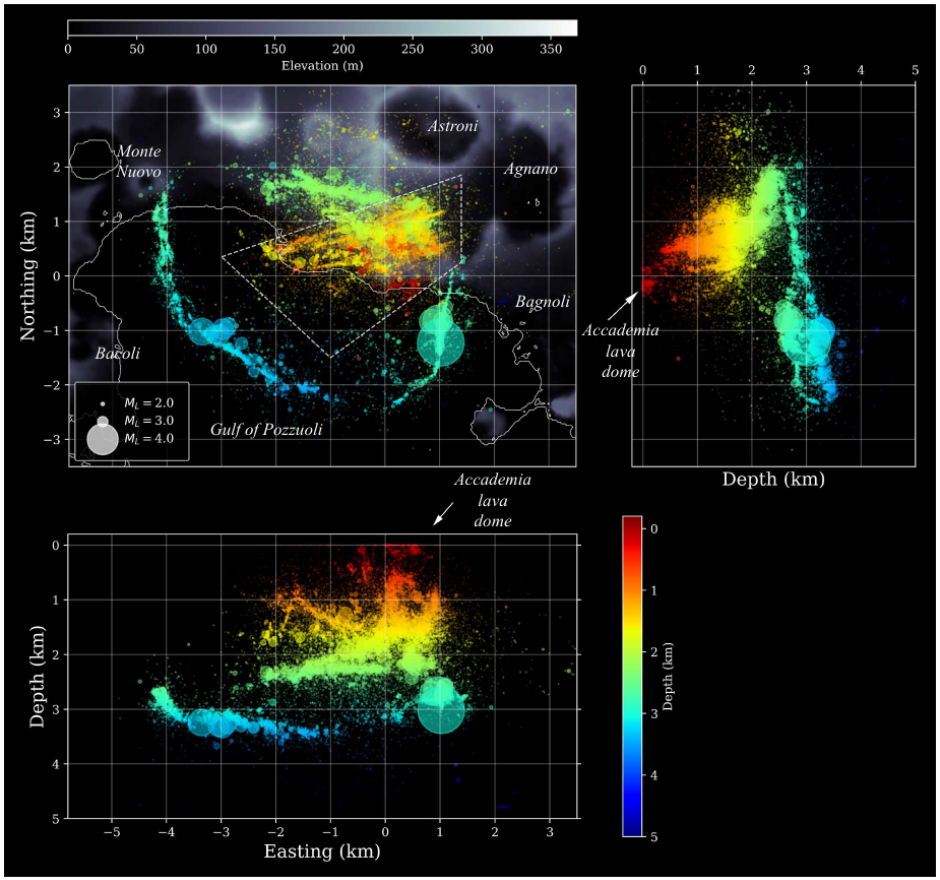
Independent result

All panels/views — not reproduced

Panels A and B need multiyear calibrated/relocated events and magnitudes. The current 740 initial associations remain provisional.

Figure 3: Spatial distribution of seismicity

Published reference



Tan et al. (2025), tanmodel.pdf, p. 4.
DOI: 10.1126/science.adw9038.

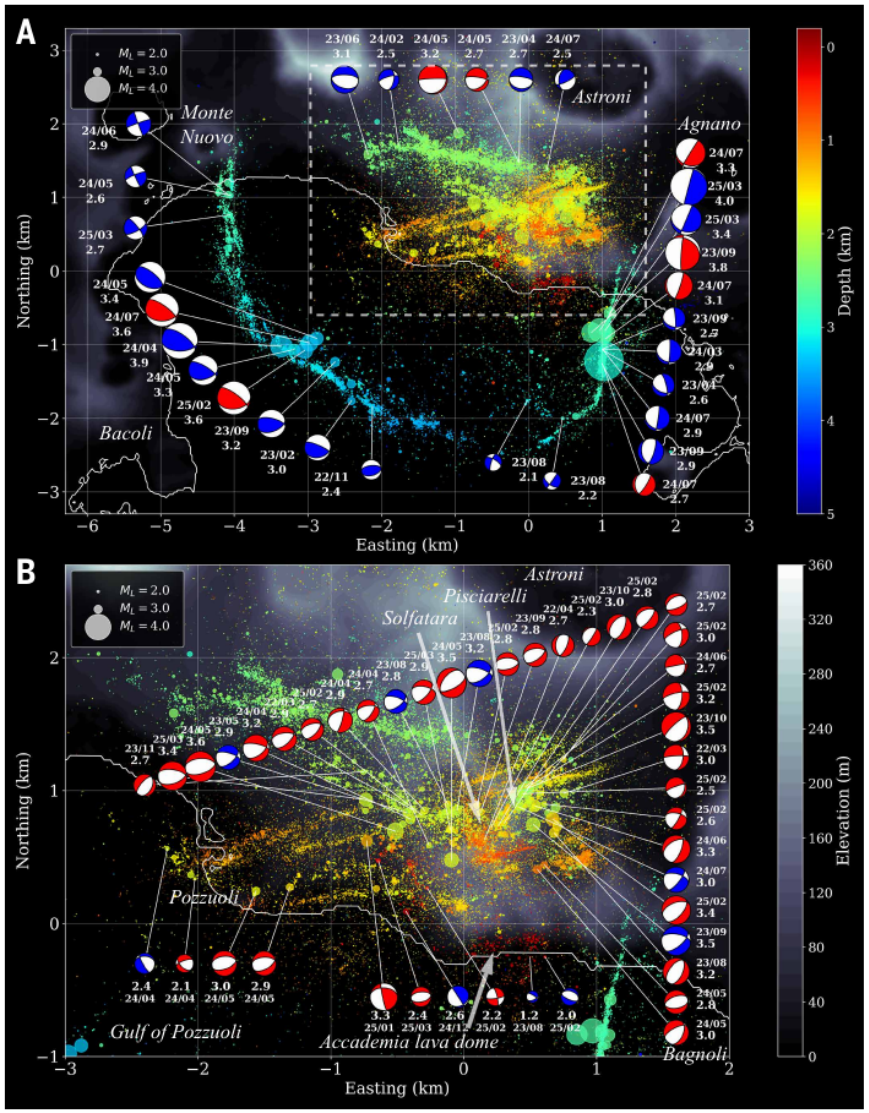
Independent result

All panels/views — not reproduced

The map and both cross sections require independently calibrated and double-difference-relocated events. Provisional diagnostic locations are not substitutes.

Figure 4: Focal mechanisms of larger events

Published reference



Tan et al. (2025), tanmodel.pdf, p. 5.
DOI: 10.1126/science.adw9038.

Independent result

All panels/views — not reproduced

Panels A and B require independent SKHASH/gCAP solutions and reliable locations. No focal inversion was executed.

Figure S1: Recorded station availability

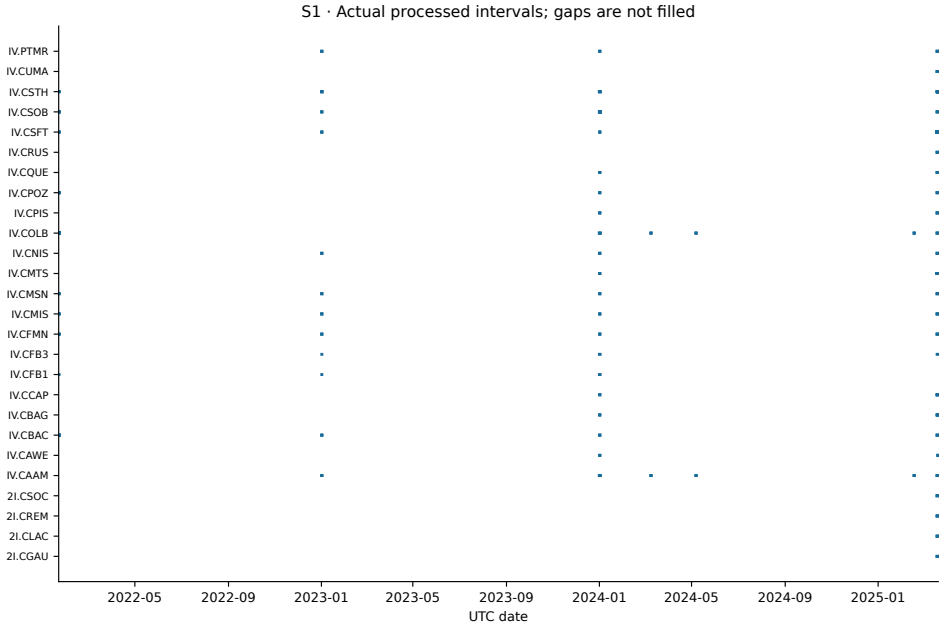
Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 12.
DOI: 10.1126/science.adw9038.

Independent result

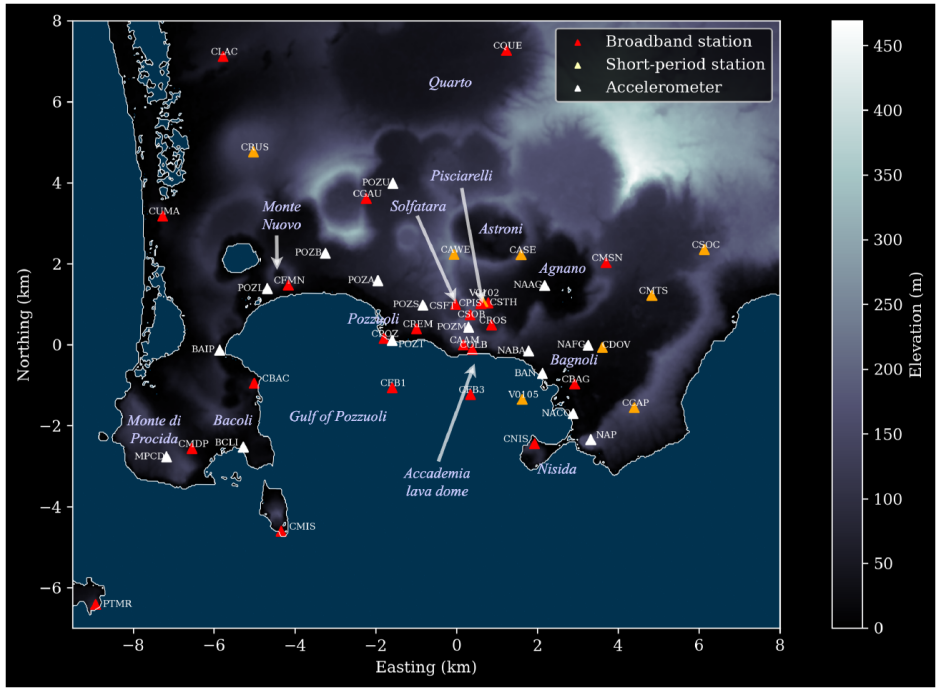
Panel S1 — partially reproduced



Partial: 318 processed files, 26 station names and 2955.824 channel-hours after union of overlapping intervals. Empty intervals are not connected or counted as coverage.

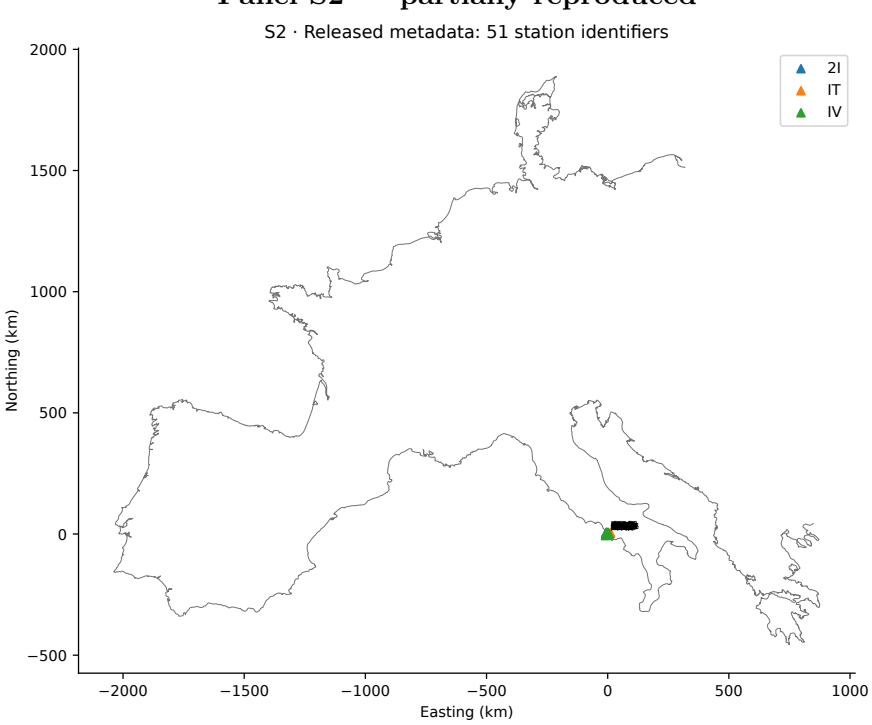
Figure S2: Station locations

Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 13.
DOI: 10.1126/science.adw9038.

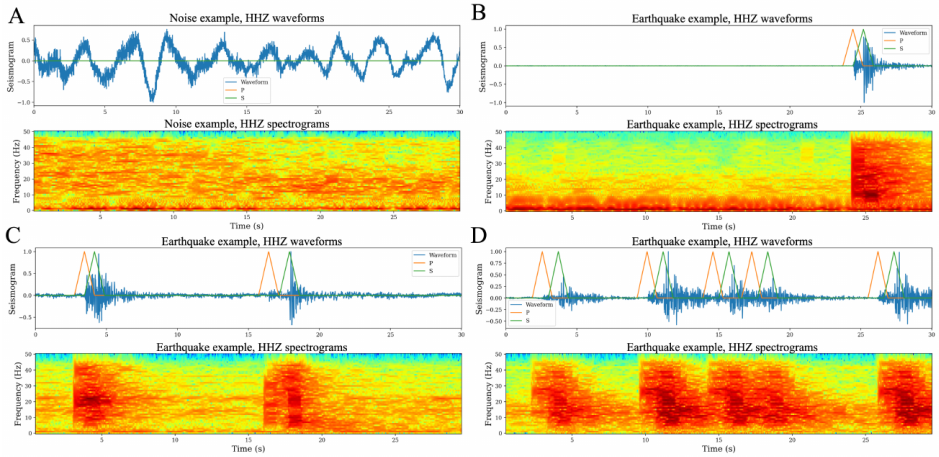
Independent result



Partial: released station coordinates and retrieved IV/2I response metadata. Exact membership of the original 48 instruments, station epochs and the basemap are not fully reconciled.

Figure S3: Training examples

Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 14.
DOI: 10.1126/science.adw9038.

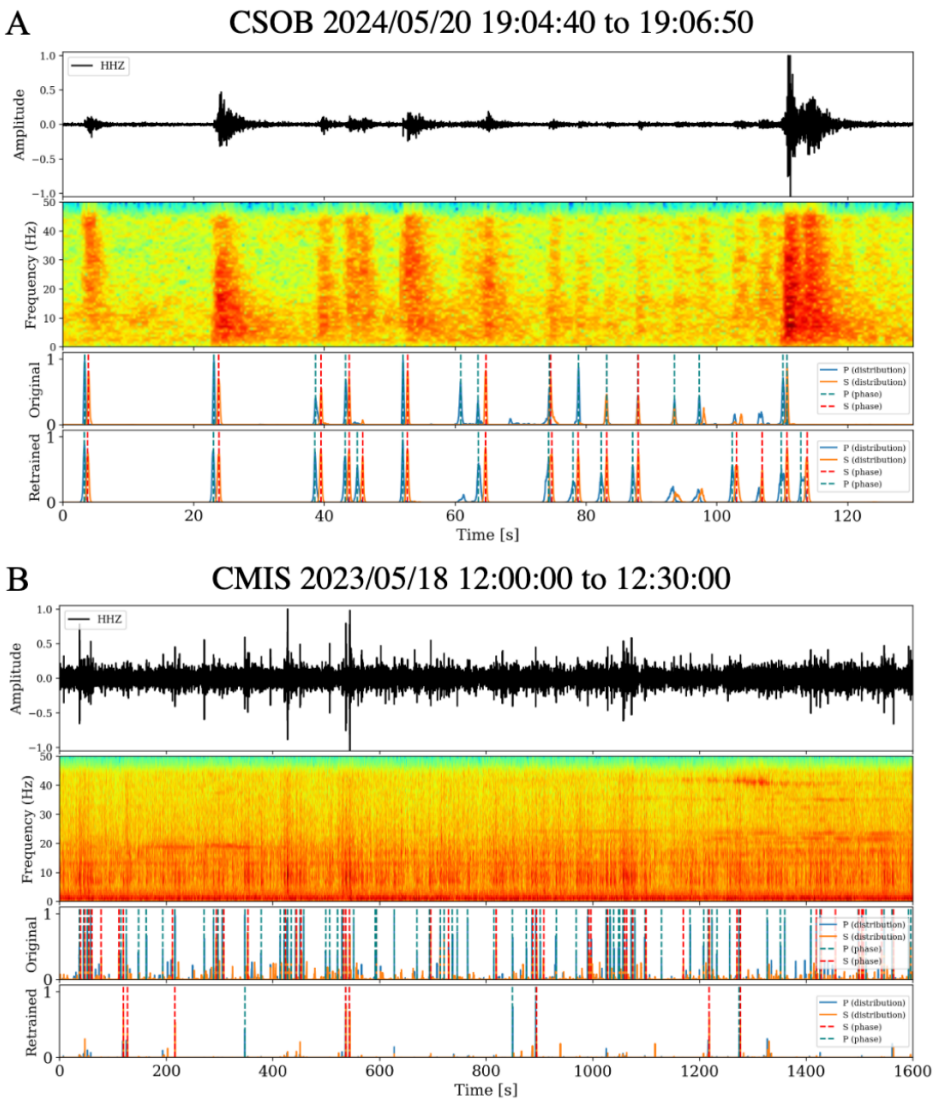
Independent result

All panels/views — not reproduced

Panels A–D require the specified training traces, annotations and augmentation examples. These were not reconstructed; downstream use of trained weights is not retraining.

Figure S4: PhaseNet benchmark examples

Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 15.
DOI: 10.1126/science.adw9038.

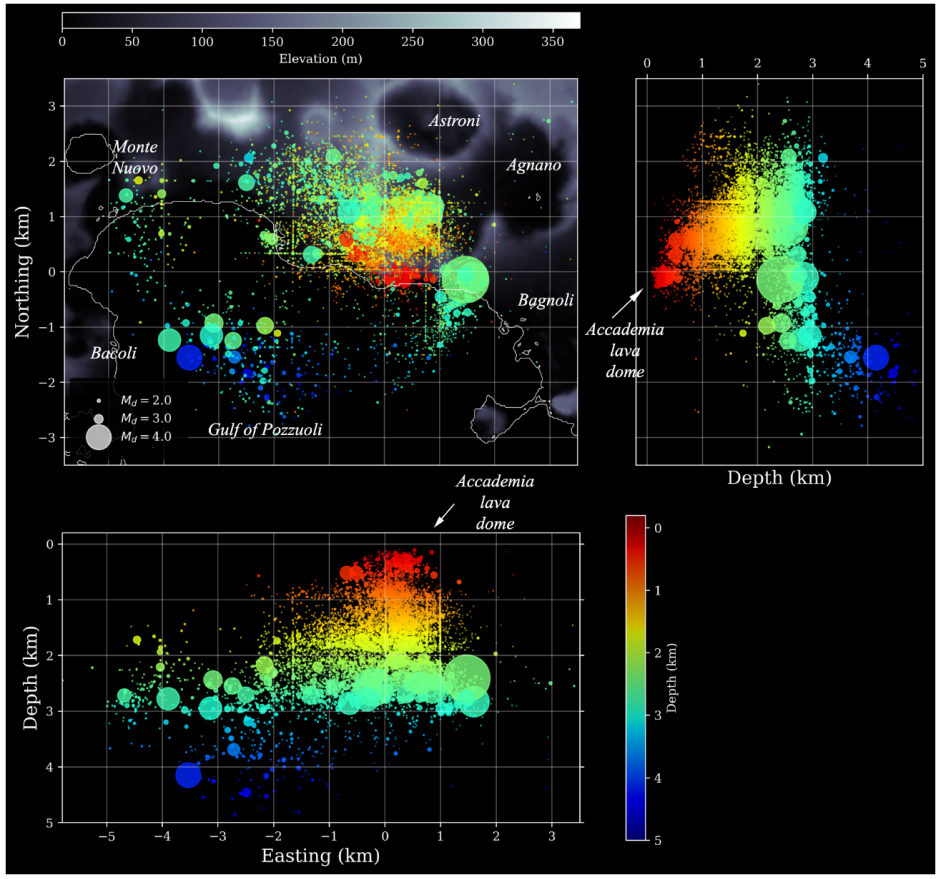
Independent result

All panels/views — not reproduced

Panels A and B were not generated for the exact CSOB swarm and CMIS noise examples. Other unlabeled inference comparisons remain diagnostics.

Figure S5: Routine-catalog locations

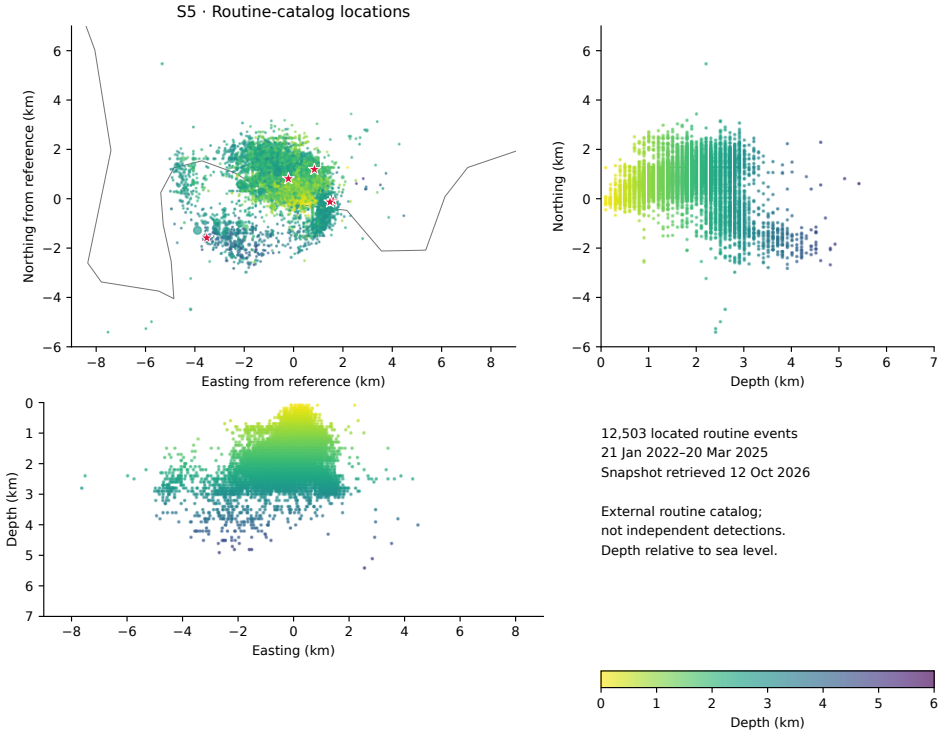
Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 16.
DOI: 10.1126/science.adw9038.

Independent result

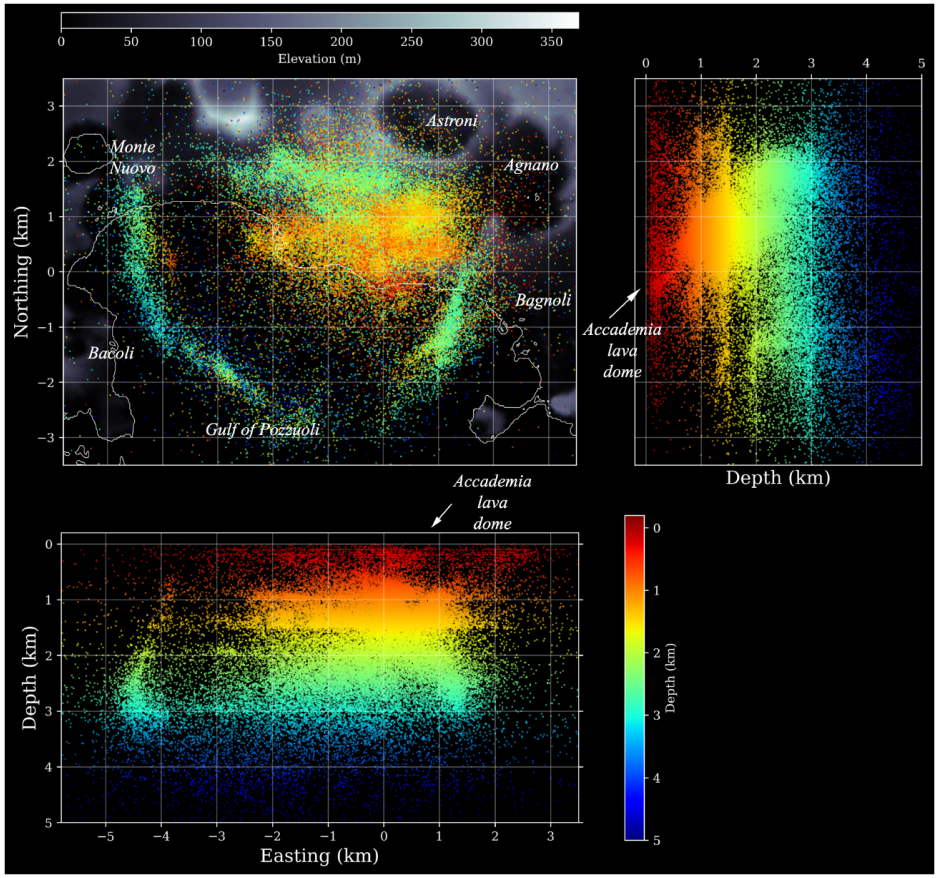
Panel S5, all views — partially reproduced



External routine-catalog data, independently selected and plotted: 12,503 located events, 21 January 2022–20 March 2025. This is 420 more than the published historical snapshot. No author-derived catalog is used to adjust that difference; these are not independent detections.

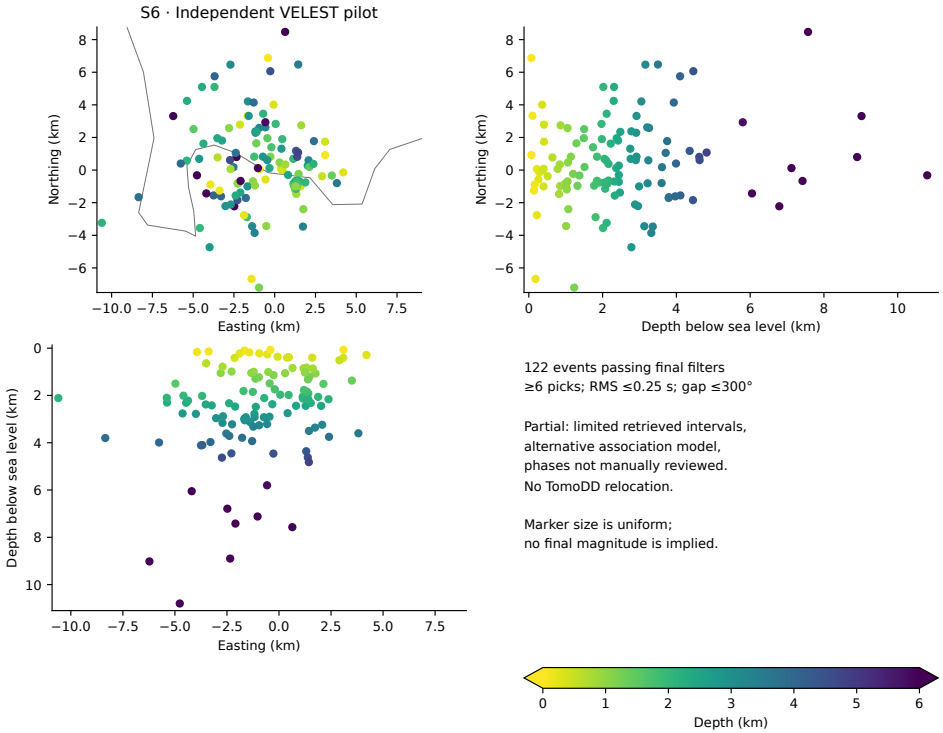
Figure S6: Initial VELEST locations

Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 17.
DOI: 10.1126/science.adw9038.

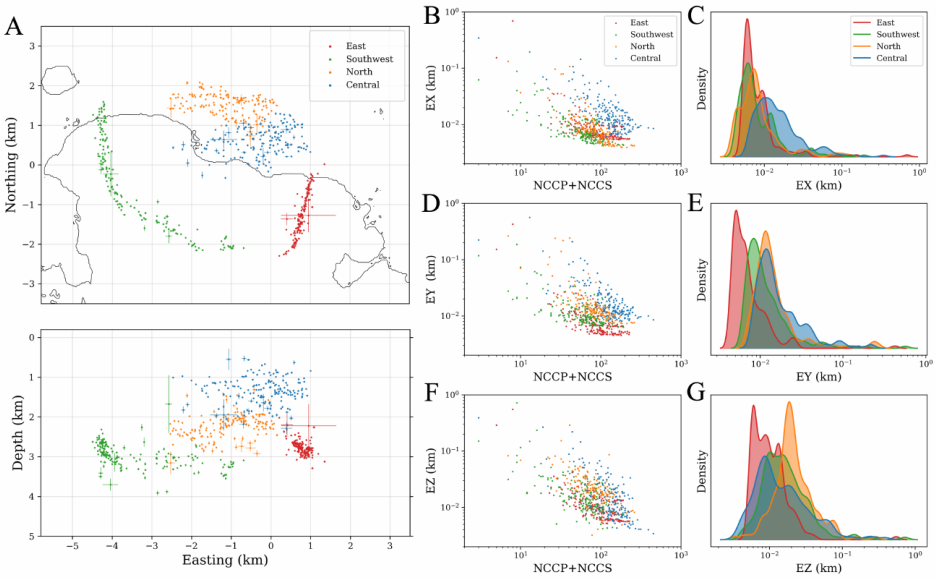
Independent result



Partial independent VELEST pilot: 122 events pass the final six-pick, 0.25 s RMS and 300-degree gap filters. This limited, unreviewed sample is not the published 61,809-event catalog. The initial association model and regional averaging choices differ or are assumed; no TomoDD refinement is implied.

Figure S7: Location uncertainties

Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 18.
DOI: 10.1126/science.adw9038.

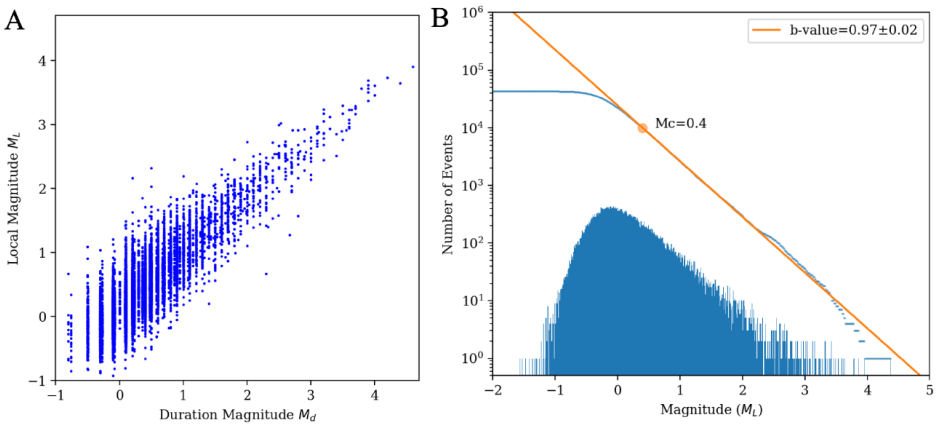
Independent result

All panels/views — not reproduced

Panels A–G require independently selected regional subsets, HypoDD SVD locations and coordinate uncertainties. None of these scientific inversions ran.

Figure S8: Magnitude calibration and frequency distribution

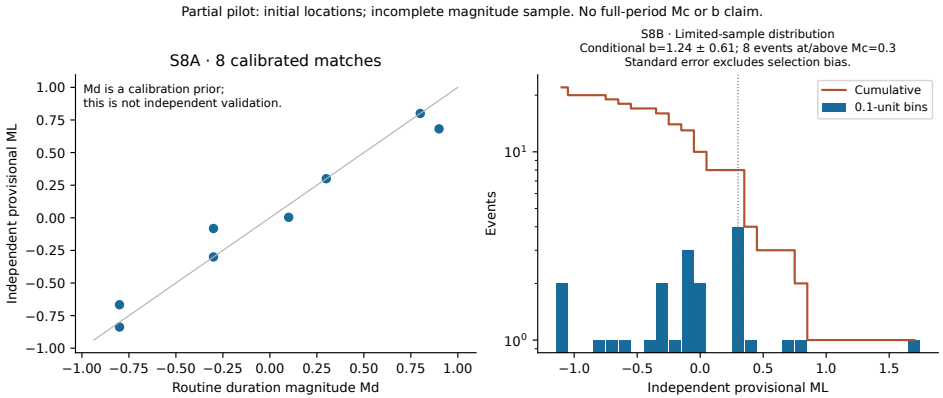
Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 19.
DOI: 10.1126/science.adw9038.

Independent result

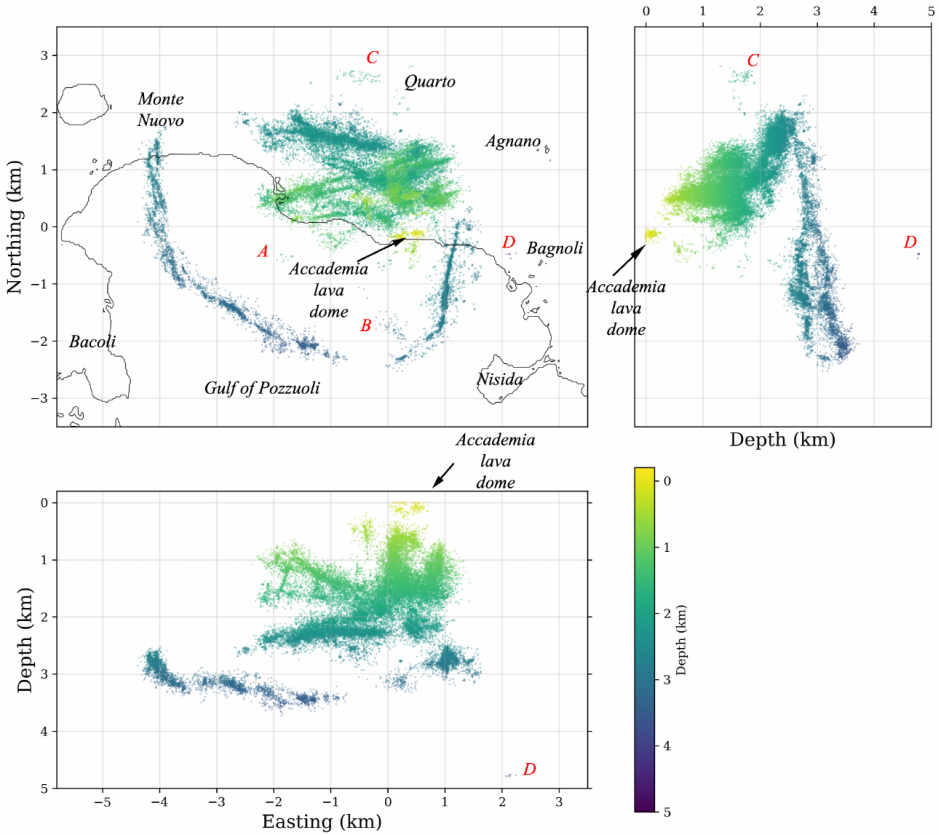
Panel S8A–B — partially reproduced



Partial amplitude-ratio pilot: 22 provisional magnitudes from independent observations, with routine Md as the documented calibration prior. A is not an independent validation of that prior. B is a limited-sample histogram with a conditional likelihood estimate, not the full-period completeness or b-value result. Locations are initial associations, before final relocation.

Figure S9: Best-located earthquake subset

Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 20.
DOI: 10.1126/science.adw9038.

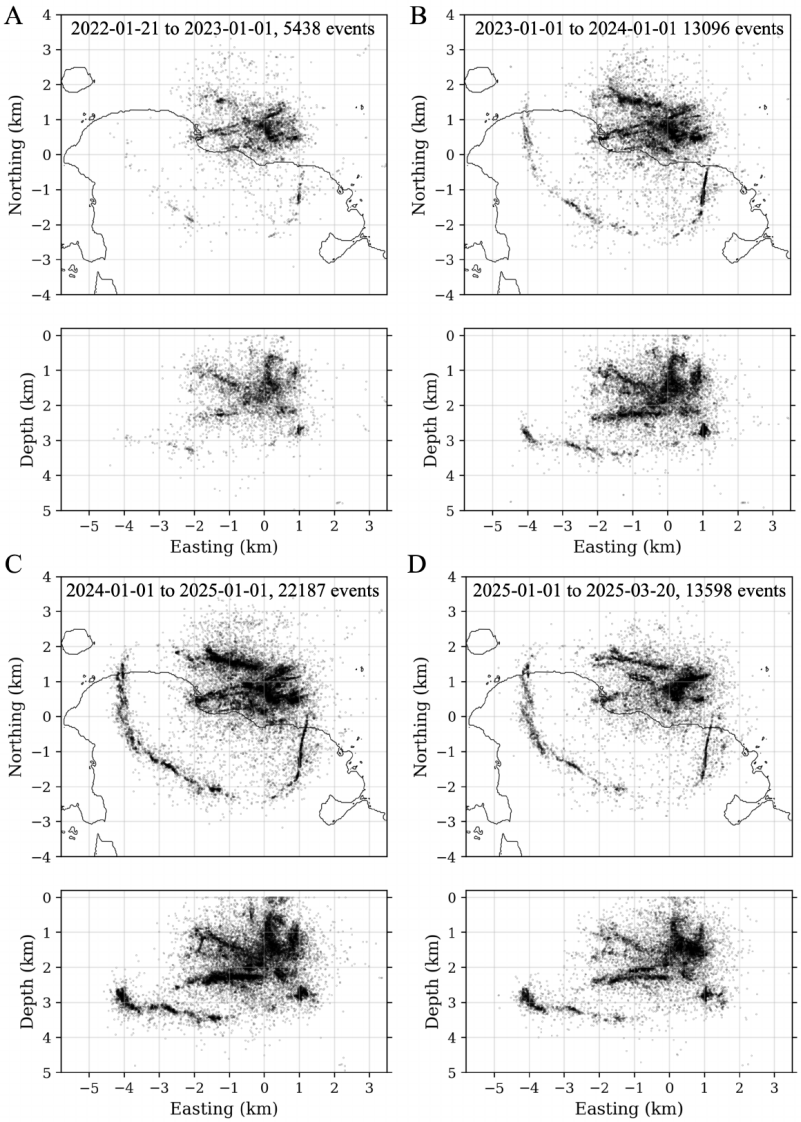
Independent result

All panels/views — not reproduced

All three views require independently relocated events and cross-correlation observation counts for the published quality filter. These outputs are absent.

Figure S10: Seismicity by year

Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 21.
DOI: 10.1126/science.adw9038.

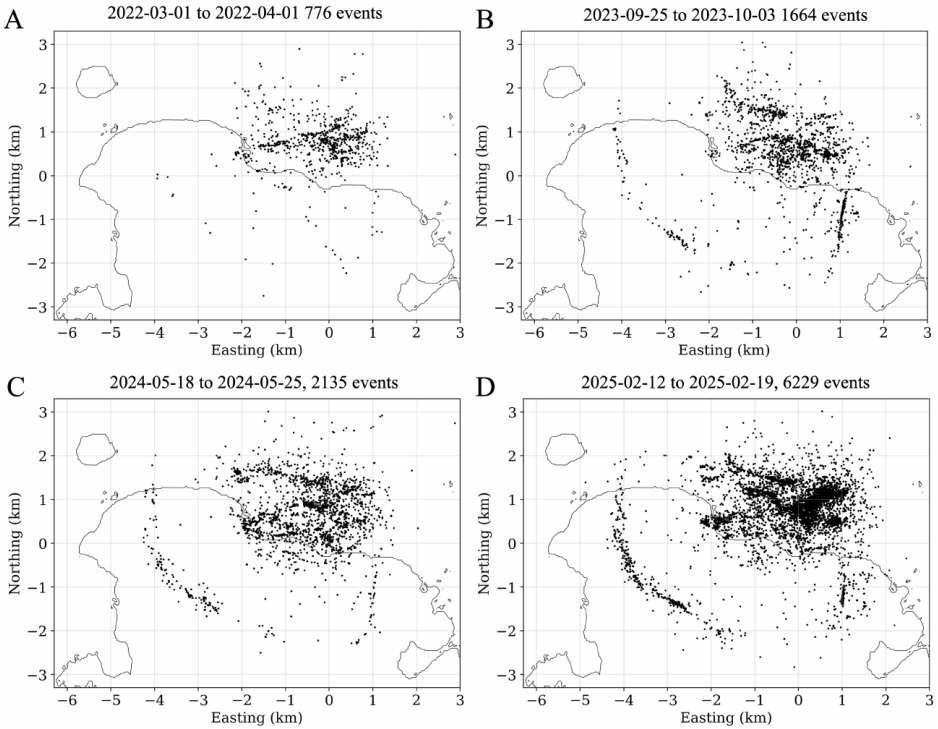
Independent result

All panels/views — not reproduced

The four annual groups for 2022–2025 require the independent multi-year relocated catalog. The available sample does not support them.

Figure S11: Four major swarms

Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 22.
DOI: 10.1126/science.adw9038.

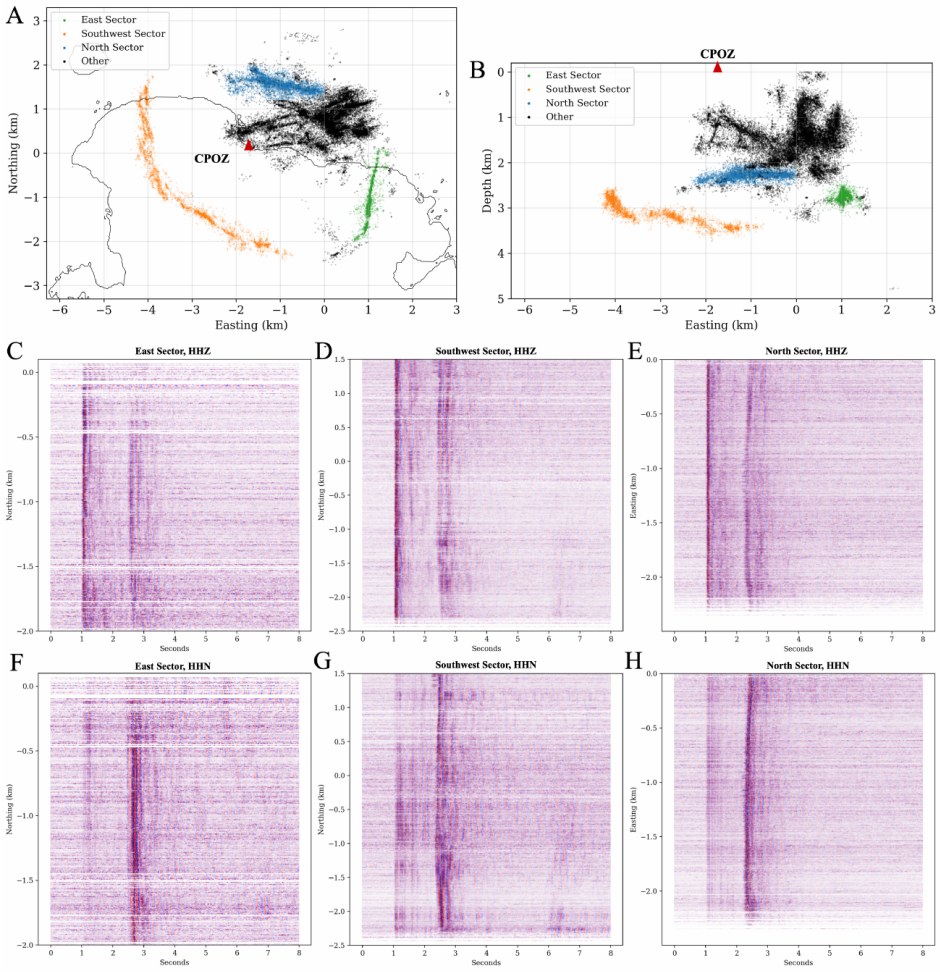
Independent result

All panels/views — not reproduced

The four swarm maps require the independent relocated catalog and justified swarm windows. These selections and locations remain incomplete.

Figure S12: Ring sectors and CPOZ common-receiver gathers

Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 23.
DOI: 10.1126/science.adw9038.

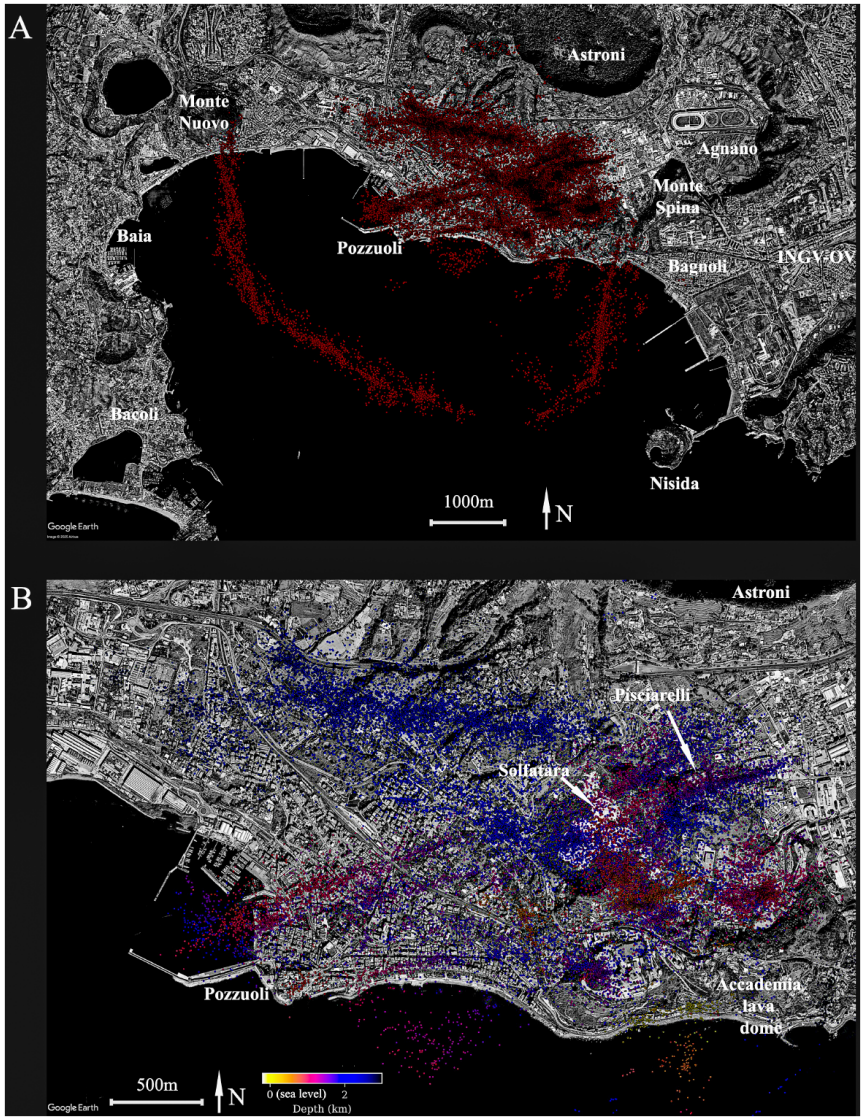
Independent result

All panels/views — not reproduced

Panels A–H need relocated sector selections and manual depth review. 40 independent CPOZ P arrivals now exist, but the required sector gathers remain unfinished.

Figure S13: Relocated seismicity on satellite imagery

Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 24.
DOI: 10.1126/science.adw9038.

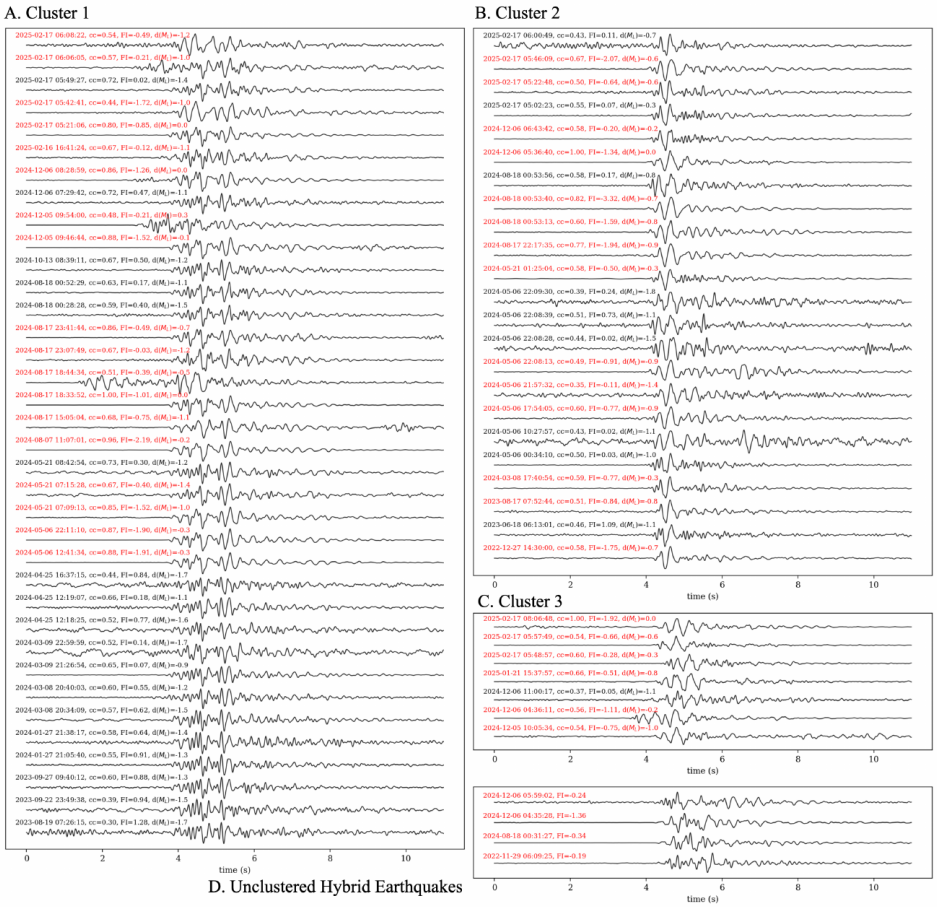
Independent result

All panels/views — not reproduced

Panels A and B require reliable independent relocated coordinates and an appropriately sourced basemap. The relocated catalog does not exist.

Figure S14: Hybrid-event clusters

Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 25.
DOI: 10.1126/science.adw9038.

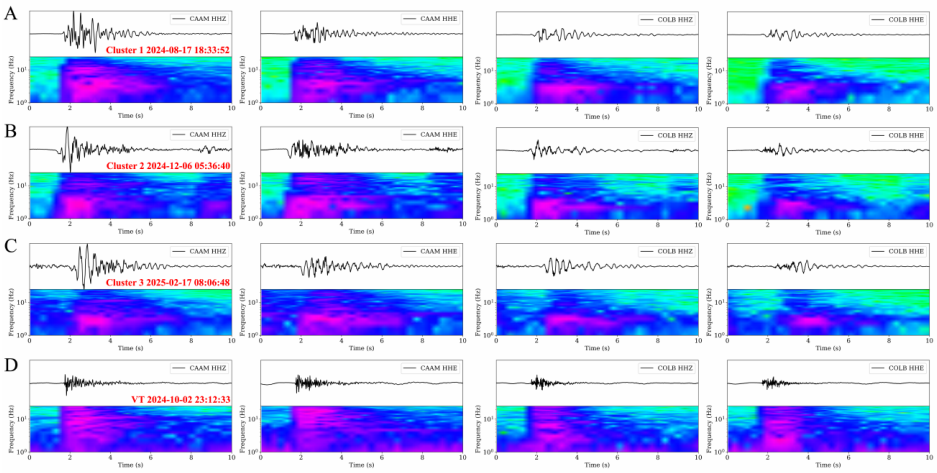
Independent result

All panels/views — not reproduced

The cluster histories require three-year exposure, confirmed events and relative magnitudes. The present 59.757 hour scan produced 702 unconfirmed threshold crossings, not reviewed families.

Figure S15: Hybrid and shallow-event waveforms

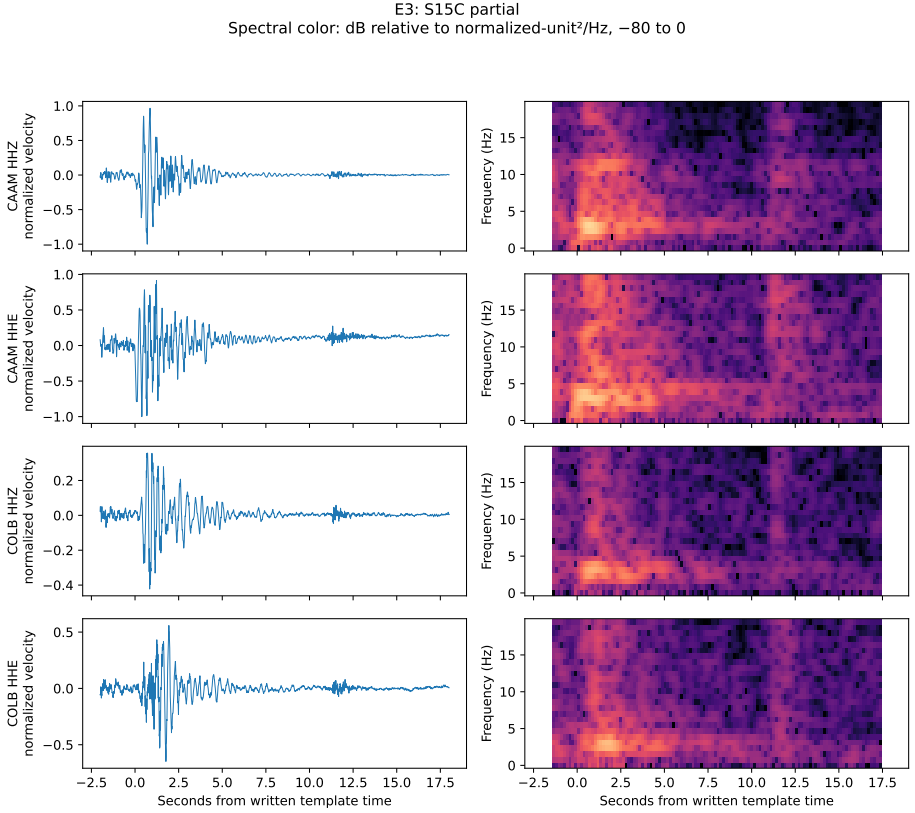
Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 26.
DOI: 10.1126/science.adw9038.

Independent result

Panel S15C — partially reproduced



Panel C is partial: independently processed E3 CAAM/COLB observations. Onset alignment, display filtering and spectral settings differ or are incompletely specified. A, B and D have placeholders rather than substitute waveforms.

Panels A, B and D — not reproduced

Panels A and B use different published events from the saved E1/E2 templates; D has no independently selected matching example. Those panels remain not reproduced.